Rafail Psyroukis - Curriculum Vitae

Contact Details

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Research Interests

Number Theory, Modular Forms of Higher Rank (Siegel, Hermitian, Orthogonal), Algebraic Modular Forms, L-functions, Fourier-Jacobi Forms, Quadratic Forms, Clifford Algebras, Quaternion Algebras.

Employment

University of Primorska — Koper, Slovenia
Assistant with PhD (Postdoctoral Researcher). — 09/2026 - Present.

Education

Durham University — Durham, UK
PhD in Mathematics. — 10/2021 - 01/2026

- Supervisor: Prof. Thanasis Bouganis.
- Thesis Title: On Fourier–Jacobi Dirichlet Series for Hermitian and Orthogonal Modular Forms.

University of Cambridge, Churchill College — Cambridge, UK
Master of Mathematics ("Cambridge Part III"). — 10/2020 - 07/2021

- Grade: Distinction, Rank: 29/272.
- Part III Essay (equivalent to a Master's Thesis): Two-Descent on the Jacobian of a Hyperelliptic Curve.
- Supervisor: Prof. Tom Fisher.

University of Cambridge, Churchill College — Cambridge, UK
B.A. (Honours) Mathematics. — 10/2017 - 07/2020

- Grade: First Class Honours, Rank: N/A (due to COVID-19).

Publications and Preprints

1. A Fourier-Jacobi Dirichlet series attached to modular forms of $\mathrm{SO}(2,4)$ (with T. Bouganis). *Submitted.* [arXiv].
2. On a Rankin-Selberg integral of three Hermitian cusp forms (with T. Bouganis). ***Ann. Math. Québec**, 50, 633-693 (2026).* [Journal], [arXiv].
3. A note on the twisted degree 6 L-function for Hermitian cusp forms of degree 2. ***Int. J. Number Theory**, 22, 06 (2026).* [Journal], [arXiv].
4. Analytic properties of an orthogonal Fourier-Jacobi Dirichlet series. ***Ramanujan J**, 69, 94 (2026).* [Journal], [arXiv].
5. A Fourier-Jacobi Dirichlet series for cusp forms on orthogonal groups. ***Res. number theory** 11, 90 (2025).* [Journal], [arXiv].

Teaching/ Marking/ Mentoring

University of Durham.

Number Theory III | Durham University — 2025 - 2026

- Marker for third year module "Number Theory III", Durham University.
- Marked weekly assignments.

Discrete Mathematics | Durham University 2025 - 2026

- Marker for first year module "Discrete Mathematics", Durham University.
- Marked weekly assignments.

Algebra II | Durham University 2022 - 2025

- Tutor and marker for the second year module "Algebra II", Durham University.
- Provided bi-weekly tutorials to groups of roughly 16 students and marked weekly assignments.

Elementary Number Theory II | Durham University 2023 - 2024

- Tutor and marker for the second year module "Elementary Number Theory II", Durham University.
- Provided bi-weekly tutorials to groups of roughly 16 students and marked weekly assignments.

Linear Algebra | Durham University 2021 - 2023

- Marker for first year module "Linear Algebra", Durham University.
- Marked weekly assignments.

Other Activities.

Marker | Cambridge University Press & Assessment 06/2026 - 07/2026

- Marked STEP 2 and STEP 3 Exam papers.

Problem Solving Matters | Durham University 06/2024 - 09/2024

- Mentor for students preparing for mathematics admissions tests such as STEP/MAT/TMUA.
- Program run by MEI Advanced Mathematics Support Programme.

Talks Given

Building Bridges, Będlewo Palace. 18/08/2026
"A Fourier-Jacobi Dirichlet series attached to modular forms of $\mathrm{SO}(2,4)$*."*

Panhellenic Conference of Mathematical Analysis, NTUA. 21/12/2025
*"Analytic properties of a certain Dirichlet series attached to orthogonal modular forms." (**Invited**)*

33èmes Journées Arithmétiques, University of Luxembourg. 01/07/2025
"A Dirichlet series attached to orthogonal modular forms".

PGR Research Day, Durham University. 12/05/2025
"From classical to Siegel modular forms".

Number Theory Seminar, Adam Mickiewicz University. 04/11/2024
*"Analytic properties of Fourier-Jacobi Dirichlet series". (**Invited**)*

Willmore Pure Postgraduate Day, Durham University. 08/10/2024
*"Analytic Properties of Rankin-Selberg Convolutions". (**Invited**)*

Y-RANT VI, University of Oxford. 01/08/2024
"A Dirichlet series for modular forms on orthogonal groups".

British Mathematical Colloquium, University of Manchester. 19/06/2024
"A Dirichlet Series attached to orthogonal modular forms".

PGR Research Day, Durham University. 22/05/2024
"Modular Forms and Applications".

Arithmetic Study Group, Durham University. 30/01/2024
*"On a Rankin-Selberg integral of three Hermitian cusp forms". (**Invited**)*

GANDALF Postgraduate Seminar, Durham University. 06/12/2023
"Hermitian Modular Forms and Dirichlet Series".

Number Theory Seminar, University of Nottingham. 01/11/2023
*"On a Rankin-Selberg integral of three Hermitian cusp forms". (**Invited**)*

Y-RANT V, University of Cambridge. 07/09/2023
"On a Rankin-Selberg integral of three Hermitian cusp forms and its relation with the twisted spinor L-function".

Conferences and Workshops

Building Bridges. 10th - 21st August 2026
Summer School and Workshop on Automorphic Forms and Related Topics.

Panhellenic Conference of Mathematical Analysis. 18th - 20th Dec. 2025
National Technical University of Athens.

Heilbronn Annual Conference 2025. 10th - 11th July 2025
University of Bristol.

33èmes Journées Arithmétiques. 30th June - 4th July 2025
University of Luxembourg.

British Mathematical Colloquium. 17th - 20th June 2024
University of Manchester.

1st Greek Number Theory Meeting. 21st - 22nd Dec. 2023
National and Kapodistrian University of Athens.

Automorphic Forms and L-functions of higher rank. 11th - 15th Sept. 2023
Queen Mary University London.

Arithmetic of L-functions. 22nd - 26th May 2023
Instituto de Ciencias Matemáticas.

Northern Number Theory Seminar (NuTS). 11th May 2023
University of York.

Heilbronn Doctoral Partnership Summer Workshop. July - August 2022
Group work on the project "Latency in Communications", proposed by the DSTL.

Scholarships

Doctoral Research Excellence Award. 2025
Awarded annually to a PhD student who has demonstrated excellence in research throughout their doctoral studies in Mathematical Sciences, Durham University.

Heilbronn Doctoral Partnership Studentship. 2021 - 2025
Studentship offered by the Heilbronn Institute for Mathematical Research in support of my PhD studies, grant number (EP/V521917/1).

Onassis Foundation Scholarship. 2022 - 2025
Scholarship offered to academically excellent students with Greek citizenship, in support of their research.

Cambridge Bursary. 2017 - 2021

Income-assessed bursary, maximum value offered each year.

Scholarship for Academic Performance. 2019 - 2021

Scholarship awarded to students who achieve a First (undergraduate studies) or a Distinction (master's studies) from Churchill College, University of Cambridge.

Professional Service

Referee for: Journal of Number Theory, International Journal of Number Theory.

Reviewer for: Mathematical Reviews.

Awards and Honours

- **International Mathematical Olympiad (IMO)**, Bronze Medal. July 2017
- **Balkan Mathematical Olympiad (BMO)**, Bronze Medal. May 2017
- **Mediterranean Mathematical Olympiad**, Honourable Mention. April 2017
- **Greek Physics Contest**, 24th prize. May 2017
- **Greek Mathematical Olympiad**, Silver Medal (×4). Feb 2014 - 2017

Skills

Languages: English, Greek (Native).

Programming: LaTeX, MATLAB, Python, SageMath, MAGMA.